Author

Nikhil Sharma
24F1002035
24f1002035@ds.study.iitm.ac.in
I am a second-year student enrolled in the IIT Madras BS in Data Science program. I have a strong interest in Artificial Intelligence and the wide range of technologies that surround and support it.

Description

This project is a web-based multi-user 4-wheeler vehicle parking system built with Flask. It allows admins to manage parking lots and users to book and release parking spots in real time. The system ensures role-based access and provides dashboards for both user types. AI/LLM was used up to 10 percent in HTML and up to 7 percent in Bootstrap/CSS. LLM tools were used to style the pages

Technologies used

Flask**:** Core backend framework used to build the web application and handle routing and logic.

Jinja2**:** For rendering dynamic HTML templates using data passed from the backend.

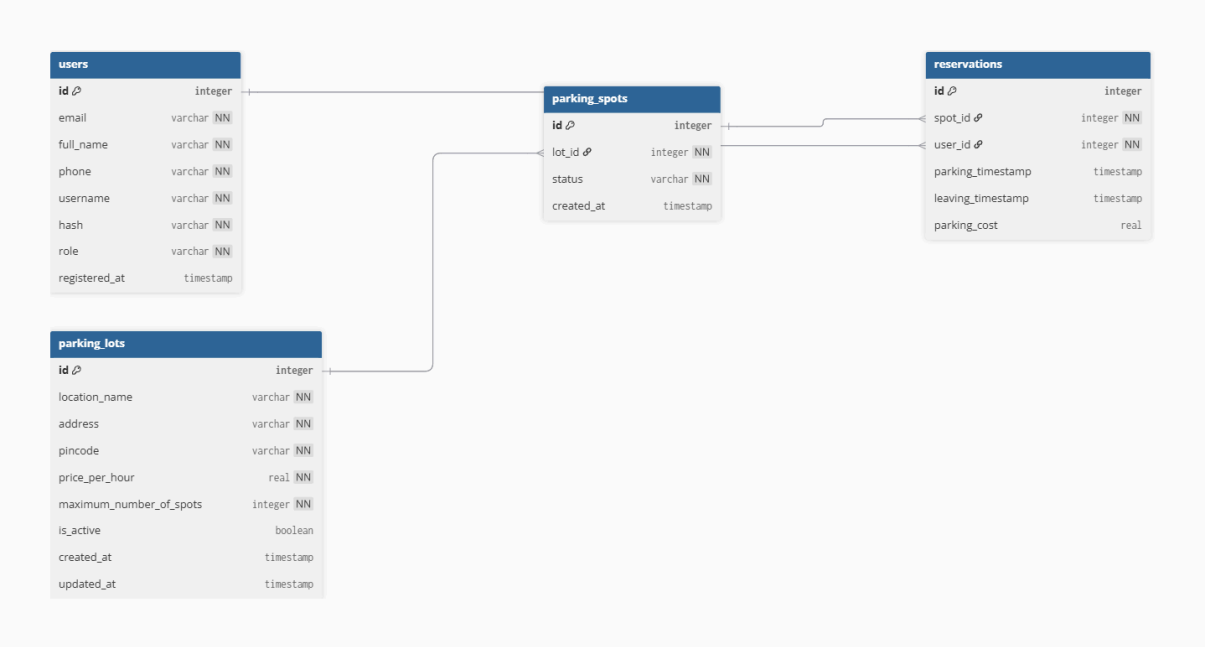
Flask-Session**:** To manage user sessions on the server-side, ensuring login persistence across pages.

SQLite**:** Lightweight, file-based database used to store user, parking lot, and reservation data.

Hashlib (Python standard library): Used to securely hash user passwords before storing them in the database.

datetime (Python standard library): For handling timestamps related to reservations and parking durations.

DB Schema Design



The tables are linked using foreign keys to maintain logical relationships between entities such as users, parking spots, and reservations. `ON DELETE CASCADE` is used to ensure data integrity by automatically removing dependent records when a parent record is deleted

Architecture and Features

The project follows a modular structure using Flask as the backend framework. Controller logics are defined in the controllers/ for both admin and user functionalities. The HTML templates are stored in the `templates/` directory and styled using Bootstrap via the `static/` directory, which holds CSS. SQLite3 is used as the backend database, and the schema is created programmatically. Sessions are used to handle login states for users and the admin.

Key features implemented include user registration and login, admin dashboard for managing parking lots and monitoring parking spots, and user functionality to reserve and release parking spots. The system automatically assigns the first available spot and calculates parking costs based on duration.

Video

https://drive.google.com/file/d/1qCMiKeaQlj6bwAJtR4KXYfut3yZhMFkO/view?usp=sharing